Nghia Trung Ngo

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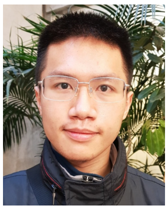

My research addresses the challenges related to learning in generalized settings, such as domain adaptation, multilingual, and multi-task learning, with applications in Information Retrieval and Extraction problems. Following the recent breakthrough of large language models (LLMs), I have participated in the development of large-scale datasets for the evaluation of LLMs' multilingual capabilities. My current works aim to further expand the current benchmarks to specialized and practical settings. By scaling the difficulty and complexity of these benchmark together with future LLM improvements, my objective is to create more robust and practical AI systems for real-world applications.

Education

2022 - 2026

Ph.D., University of Oregon in Computer Science, March 2026.
Advisor: Prof. Thien Huu Nguyen.
Thesis title: *Multidimensional Transfer Learning in Natural Language Processing: Domain, Task, Language, and Expertise Adaptation*
A two-phase investigation of transfer learning in NLP, covering adaptive transfer across domains, tasks, and languages and specialization transfer for refining large language models toward reliable deployment in high-stakes expert domains.

2015 - 2021

B.S., Hanoi University of Science and Technology (HUST) in Information Technology.
Advisor: Dr. Linh Ngo Van.
Thesis title: *Unsupervised Domain Adaptation for Event Detection*
Proposed an Unsupervised Domain Adaptation framework for Event Detection task by utilizing domain-specific adapters and optimal transport-based adversarial learning.

Employment History

2022 - 2026 **Research Assistant,** University of Oregon.
+ *Research Topics: Transfer Learning, Multilingual and Multi-Task Learning*

- Proposed a novel framework for joint learning of multiple Information Extraction tasks in Domain Adaptation setting.
- Introduced a generalized multi-transfer setting for cross-lingual learning and proposed an effective graph relational-transfer method to address the problem.

+ *Research Directions: Retrieval-Augmented Generation (RAG), LLM Benchmarking*
Working on scaling and expanding the difficulty and complexity of current benchmarks. In particular:

- Medical Retrieval: Developed a Medical RAG Benchmark that incorporates practical scenarios with increasing amounts of noise and misinformation.
- Multilingual Integration: Benchmarking multilingual RAG system's ability to integrate diverse knowledge from a growing number of cultural sources.
- Commonsense Reasoning: Exploring LLM's reasoning patterns with progressively more complex commonsense questions.

2025 - 2026 **Teaching Assistant,** University of Oregon.
Supported instruction for CS 313 and CS 315 in data structures and algorithms.

2020 - 2021 **NLP Research Residence,** VinAI Research.
+ *Research Topics: Domain Adaptation and Event Extraction*

- Explored methods for Domain Adaptation problem
- Proposed an efficient adapter-based representation learning framework for effective Unsupervised Domain Adaptation for Event Detection task.

+ *Applied project: Benchmarking Vision Transformer for Fisheye Images*
Benchmarked various vision transformer architectures' capacity to process fisheye images for autonomous driving applications.

2018 - 2019 **Research Student,** Data Science Laboratory - HUST.
+ *Research Topics: Machine Learning and Topic Modeling*
Fundamental training on machine learning theory and probabilistic topic modeling problems.

Research Publications

2025 **mSCoRe: a Multilingual and Scalable Benchmark for Skill-based Commonsense Reasoning**
Nghia Trung Ngo, Franck Dernoncourt, Thien Huu Nguyen
Proceedings of LREC 2026

Comprehensive and Practical Evaluation of Retrieval-Augmented Generation Systems for Medical Question Answering
Nghia Trung Ngo, Chien Van Nguyen, Franck Dernoncourt, Thien Huu Nguyen
AAAI 2026 Workshop on AI for Scientific Research

Research Publications (continued)

ULLME: A Unified Framework for Large Language Model Embeddings with Generation-Augmented Learning
Hieu Man, **Nghia Trung Ngo**, Franck Dernoncourt, Thien Huu Nguyen
Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing: System Demonstrations, EMNLP 2024

Hierarchical Selection of Important Context for Generative Event Causality Identification with Optimal Transports
Hieu Man, Chien Van Nguyen, **Nghia Trung Ngo**, Linh Ngo, Franck Dernoncourt, Thien Huu Nguyen
Proceedings of the 2024 Joint International Conference on Computational Linguistics, Language Resources and Evaluation, LREC-COLING 2024

2023 **Zero-shot Cross-lingual Transfer Learning with Multiple Source and Target Languages for Information Extraction: Language Selection and Adversarial Training**
Nghia Trung Ngo, Thien Huu Nguyen
arXiv preprint arXiv:2411.08785

Culturax: A cleaned, enormous, and multilingual dataset for large language models in 167 languages
Thuat Nguyen, Chien Van Nguyen, Viet Dac Lai, Hieu Man, **Nghia Trung Ngo**, Franck Dernoncourt, Ryan A Rossi, Thien Huu Nguyen
Proceedings of the 2024 Joint International Conference on Computational Linguistics, Language Resources and Evaluation, LREC-COLING 2024

Okapi: Instruction-tuned large language models in multiple languages with reinforcement learning from human feedback
Viet Dac Lai, Chien Van Nguyen, **Nghia Trung Ngo**, Thuat Nguyen, Franck Dernoncourt, Ryan A Rossi, Thien Huu Nguyen
Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing: System Demonstrations, EMNLP 2023

Chatgpt beyond english: Towards a comprehensive evaluation of large language models in multilingual learning
Viet Dac Lai, **Nghia Trung Ngo**, Amir Pouran Ben Veyseh, Hieu Man, Franck Dernoncourt, Trung Bui, Thien Huu Nguyen
Findings of the Association for Computational Linguistics, EMNLP 2023

2022 **Unsupervised domain adaptation for joint information extraction**
Nghia Trung Ngo, Bonan Min, Thien Nguyen
Findings of the Association for Computational Linguistics, EMNLP 2022

Unsupervised domain adaptation for text classification via meta self-paced learning
Nghia Trung Ngo, Linh Van Ngo, Thien Huu Nguyen
Proceedings of the 29th International Conference on Computational Linguistics, COLING 2022

Selecting optimal context sentences for event-event relation extraction
Hieu Man, **Nghia Trung Ngo**, Linh Ngo Van, Thien Huu Nguyen
Proceedings of the AAAI conference on artificial intelligence, AAAI 2022

FAMIE: A fast active learning framework for multilingual information extraction
Minh Van Nguyen, **Nghia Trung Ngo**, Bonan Min, Thien Huu Nguyen
Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies: System Demonstrations, NAACL 2022

Research Publications (continued)

2021 **Unsupervised domain adaptation for event detection using domain-specific adapters**
Nghia Ngo Trung, Duy Phung, Thien Huu Nguyen
Findings of the Association for Computational Linguistics, ACL-IJCNLP 2021

Modeling document-level context for event detection via important context selection
Amir Pouran Ben Veyseh, Minh Van Nguyen, **Nghia Ngo Trung**, Bonan Min, Thien Huu Nguyen
Proceedings of the 2021 conference on empirical methods in natural language processing, EMNLP 2021

2020 **Learning to select important context words for event detection**
Nghia Trung Ngo, Tuan Ngo Nguyen, Thien Huu Nguyen
Advances in Knowledge Discovery and Data Mining: 24th Pacific-Asia Conference, PAKDD 2020

Services

Selected **Reviewer,** ACL, NAACL, EMNLP, AAAI, ICLR, NeurIPS, COLING, COLM

2025 **Reviewer,** The 40th Annual AAAI Conference on Artificial Intelligence (**AAAI 2026**), The Second Conference on Language Modeling (**COLM 2025**)

2024 **Reviewer,** The 39th Annual AAAI Conference on Artificial Intelligence (**AAAI 2025**), The Thirteenth International Conference on Learning Representations (**ICLR 2025**)

2023 **Reviewer,** The Forty-First International Conference on Machine Learning (**ICML 2024**)

2022 **Reviewer,** The Thirty-Seventh Annual Conference on Neural Information Processing Systems (**NeurIPS 2023**)

2021 **Reviewer,** The AAAI-2022 Workshop On Video Transcript Understanding

Skills

Languages **Vietnamese**: Native
English: IELTS 7.5 - 9.0 Reading, 8.0 Listening, 6.5 Writing, 6.5 Speaking.

Programing Language **Proficient**: Python
Familiar: C, Java, SQL

Awards and Achievements

2025 **Lorry I. Lokey Interdisciplinary Science Initiative Dissertation Scholarship, University of Oregon** awarded by the Department of Computer Science.

2023 **UO's Erwin & Gertrude Juilfs Scholarship in Computer Science** for students who show exceptional promise for achievement as evidence by GPA, originality of research or other applicable criteria.